

Retrieval Augmented Tree of Thoughts

Xuemin Liu and Jie Liu()

National Key Laboratory of Intelligent Tracking and Forecasting for Infectious Diseases, Engineering Research Center of Trusted Behavior Intelligence, Ministry of Education, College of Artificial Intelligence, Nankai University, Tianjin, China
xuemin_liu2002@163.com, jliu@nankai.edu.cn

Abstract. Large language models (LLMs) are able to answer reasoning questions through generating reasoning processes, such as Chains of Thoughts (CoT) and Trees of Thoughts (ToT). To retrieve external documents effectively for knowledge-intensive multi-step question answering (QA), we propose Retrieval Augmented Tree of Thoughts (RAToT), a retrieval framework which performs retrieval and locally optimal reasoning iteratively. RAToT inspires LLM to explore diverse reasoning paths via separating retrieved documents and selects the most promising one for each step. RAToT, which uses gpt-3.5-turbo-instruct as backbone LLM, retrieves documents with less noise than IRCoT on three open-domain multi-step QA datasets, HotpotQA, 2WikiMultihopQA and MuSiQue, and further enhances the performance of LLM QA on MuSiQue. RAToT can retrieve documents wisely and flexibly to mitigate the hallucinations in open-domain multi-step QA tasks.¹

Keywords: Retrieval Augmented Generation · Tree of Thoughts · Hallucination.

1 Introduction

Large language models achieve impressive performance in various natural language processing tasks. Unfortunately, when answering some knowledge-intensive questions, especially multi-step questions, LLM probably produces hallucinations [6,13,14,27] due to the limitation of parametric knowledge.

Retrieving documents from external knowledge sources and using them as additional input context can provide LLMs with nonparametric knowledge and mitigate hallucinations [8,10,11,26]. To retrieve helpful knowledge for more complex multi-step reasoning questions, recent works propose iterative retrieval approaches, such as Self-Ask [15], ReAct [24] and IRCoT [18]. Interleaving retrieval with popular Chain-of-Thought (CoT) prompting, IRCoT retrieval, which performs retrieval augmented CoT reasoning step by step, then retrieves additional documents using newly generated reasoning step, and iterative, is a inspiring approach to improve the accuracy of CoT reasoning.

¹ Code and data are available at <https://github.com/XMLAlep/RAToT>.

However, when generating each reasoning step, IRCoT makes once-only and general decision based on all the documents retrieved so far, which may be not the optimal way of using diverse information in retrieved documents. Besides, no retrieval can absolutely avoid retrieving noisy documents. It could be difficult for LLMs to focus on the relevant documents because of the disruption of the noisy ones [2,6,12,16]. Retrieval that uses the suboptimal reasoning steps as query may not perform well.

In order to enhance the efficacy of iterative retrieval using CoT reasoning steps, we propose Retrieval Augmented Tree of Thoughts (RAToT). Transforming the diversity of retrieval to the diversity of reasoning approach, RAToT guides LLM to explore diverse reasoning paths based on different retrieved documents and make prompting-based choices wisely. By modeling retrieval augmented reasoning process with a greedy search tree and choosing local optimal candidate for each reasoning step, RAToT can perform better iterative retrieval using reasoning steps as query. RAToT is inspired by the mechanism of human reasoning. For instance, to answer complex knowledge intensive question, one may first retrieve some similar information using search engine, then read and perform reasoning based on them separately, rather than directly give a general summary based on all of the retrieved information.

We evaluate the performance of RAToT on three open-domain multi-step QA datasets, HotpotQA[22], 2WikiMultihopQA [5] and MuSiQue [17], using gpt-3.5-turbo-instruct (4k context window). RAToT outperforms IRCoT by over 7 answer F1 point on MuSiQue. Besides, RAToT can filter out retrieved noisy documents, outperforming IRCoT by over 10 retrieval F1 point on each dataset. We show RAToT is a promising iterative retrieval approach, which is capable of performing retrieval and using retrieved documents flexibly and mitigating hallucinations in open-domain multi-step QA.

2 Related Work

2.1 RAG for Open-Domain Multi-Step QA

The answering of open-domain multi-step question requires multi-hop reasoning and inference, whose reasoning steps and their dependencies can be viewed as directed acyclic graph (DAG) [17]. In other words, Performing one step properly requires the outcome of its predecessors [18]. For example, to answer “When was the first establishment that McDonaldization is named after, open in the country HornDean is located?”, one needs to infer “McDonaldization is named after McDonald’s” and “The country HornDean is located is England”, then “The first establishment of McDonald’s in England is in 1974.”, which leads to the correct final answer.

Retrieval Augmented Generation (RAG) is used to mitigate LLM hallucinations in open-domain QA. One-hop retrieval, which retrieves a batch of documents using the question solely, struggles to retrieve relevant information of latent reasoning steps, which may be not mentioned in the question directly.

Recent works propose iterative retrieval frameworks, which can retrieve more relevant documents for later reasoning steps based on generated reasoning steps iteratively.

Self-Ask+Search Engine Press et al. [15] discuss the level of faithfulness hallucination [6] that LLM generates false answer even though the sub-questions have been answered correctly. They propose Self-Ask, which guides LLM to generate decomposed sub-question answering chain based on structured prompting, reaching answer of reasoning question by answering sub-questions iteratively. On this basis, they plug search engine in the reasoning framework to answer sub-questions instead of using LLM.

ReAct Yao et al. [24] frame the multi-step QA as a decision process consisting of two types of “action”, performing follow-up reasoning step and retrieving more documents from external corpus. The ReAct system performs reasoning and retrieving iteratively through Autonomous decision-making to answer the reasoning question.

IRCoT Combining Chain of Thoughts prompting [21] with Retrieval Augmented Generation, Trivedi et al. [18] propose IRCoT (Interleaved Retrieval guided by Chain-of-Thought). IRCoT performs iterative retrieval based on CoT and in turn generates CoT using retrieved documents, which can retrieve more relevant documents and thus reduce LLM hallucinations in reasoning QA substantially.

HGOT Fang et al. [4] HGOT (Hierarchical Graph of Thoughts) framework involves graph construction, search and inference. The graph construction and search steps generate a multi-layered graph of thoughts employing the divide-and-conquer strategy and retrieve documents recursively, and the inference step predicts the final answer using a refined self-consistency voting mechanism. HGOT is promising to improve LLM’s ability of handling extremely complex reasoning problems, but relies on the planning capabilities of LLMs.

2.2 Robustness of RAG

Documents retrieved from external knowledge bases may not enhance LLM performance in open-domain QA [6,20,25]. Cuconasu et al. [3] categorize knowledge documents into several types, i) relevant documents, documents that are contextually relevant to the query and contain the correct answer to the question, ii) related documents, documents that are also semantically similar to the query, but do not contain the answer, iii) irrelevant documents, other documents that neither similar to the query nor contain the answer. They show that adding related documents to the model input context significantly reduces accuracy. According to their empirical analysis, LLMs may erroneously focus on related

documents in the context due to their semantical similarity to the query and relevant documents, thus deviate from the correct answer and produce unexpected hallucinations.

Distinguishing the relevant documents from the related but noisy ones can be challenging for retrieval methods based on the semantical relevance between query and knowledge documents [7,9]. Yoran et al. [25] alleviate the negative effects of retrieved noisy documents through training an additional natural language inference model to discard noisy documents, or fine-tuning LLMs to enhance their retrieval-robustness. Wang et al. [20] reduce noisy retrieval by guiding LLMs to determine whether to perform retrieval based on their self-knowledge and avoiding unnecessary retrieval.

2.3 Chain of Thoughts

Chain of Thoughts Wei et al. [21] prompt LLMs with a few task exemplars and guide them to perform reasoning by producing humanoid step-by-step reasoning process, i.e. chain of thoughts. The chain-of-thought prompting has been shown to improve the performance of LLMs on a range of complex reasoning tasks, such as arithmetic and common sense reasoning. On this basis, Trivedi et al. [18] use retrieved knowledge documents to improve the factuality of chain-of-thought generation in commonsense reasoning QA.

Tree of Thoughts/Graph of Thoughts Following works propose reasoning paradigms that are closer to brain mechanism, such as Tree of Thoughts [23] and Graph of Thoughts [1]. They guide LLMs to perform complex reasoning by sampling multiple reasoning branches, combining different branches or pruning less promising ones, and enhancing reasoning steps through backtracking and refining. These frameworks can improve the reasoning ability of LLMs on tasks such as arithmetic and symbolic reasoning, but do not specially consider enhancing the factuality of reasoning through retrieving knowledge documents, thus tend to produce hallucinations in knowledge-intensive reasoning tasks.

Self-Consistency Chain of Thoughts Wang et al. [19] propose self-consistency, an unsupervised decode method which samples multiple chains of thoughts and chooses the most consistent answer as the final outcome. It significantly improves the accuracy of chain-of-thought prompting on both arithmetic and symbolic reasoning and commonsense reasoning tasks.

3 Retrieval Augmented Tree of Thoughts

3.1 Exploring Diverse Retrieval Augmented Reasoning Paths

In general retrieval scenarios, e.g. retrieving from Internet, one tends to retrieve a large amount of documents, though all semantically similar to the query, to

find the actually useful information. To augment LLM generation, prior works usually use retrieved documents as a whole, providing LLM with multifaceted information. In IRCOT retrieval, LLM decides each reasoning step using all documents retrieved so far, which is a simple approach to perform reasoning and may not make full use of the diversity of retrieved documents.

We assume that different retrieved documents are capable of inspiring LLM to explore diverse reasoning paths. For each reasoning step, we first retrieve K documents based on the original multi-step question or the newly generated reasoning step, and generate K following reasoning steps based on the documents separately. LLMs can serve as judges [28], thus we then prompt LLM to evaluate the quality of the alternative reasoning steps and choose the most promising, i.e. optimal one which is then used to retrieve more documents, as well as its corresponding document. The other documents are abandoned, as current reasoning steps inspires by them are suboptimal. They can be called back in the following if necessary. We form a tree of thoughts, which explores a locally optimal reasoning path. Only documents inspiring these chosen steps can be returned as search results.

3.2 Retrieval Augmented Tree of Thoughts

In order to perform wise and flexible iterative retrieval to answer the open-domain multi-step questions, we propose Retrieval Augmented Tree of Thoughts (RAToT) retriever based on IRCOT [18]. As is presented in Fig. 1, RAToT iteratively retrieves knowledge documents using the origin question or generated reasoning step, performs locally optimal reasoning, and returns only chosen documents. Then we use a LLM Generator to generate a chain of thoughts to answer the question based on documents returned by RAToT.

RAToT Retriever Retriever consists of Retrieval Module, Generation Module, Evaluation Module and Voting Module, and retrieves documents iteratively by generating retrieval augmented chain of thoughts, as shown in Fig. 1. Accepting a multi-step question as input, Retriever first retrieves K documents using Retrieval Module, then determines the first optimal reasoning step and corresponding document by calling the rest modules, and iterates the above process until the newly generated reasoning step contains “answer is ...” or the iteration time reaches threshold M .

Retrieval Module The Retrieval Module accepts the original question or newly generated reasoning step as input query and returns top- K documents based on some retrieval algorithm from an external knowledge base.

Generation Module Generation Module accepts the last K documents returned by Retrieval Module as input and generates K next reasoning steps based on the K documents separately, and returns a candidate set of these new reasoning steps and corresponding documents. To generate each alternative reasoning step, LLM

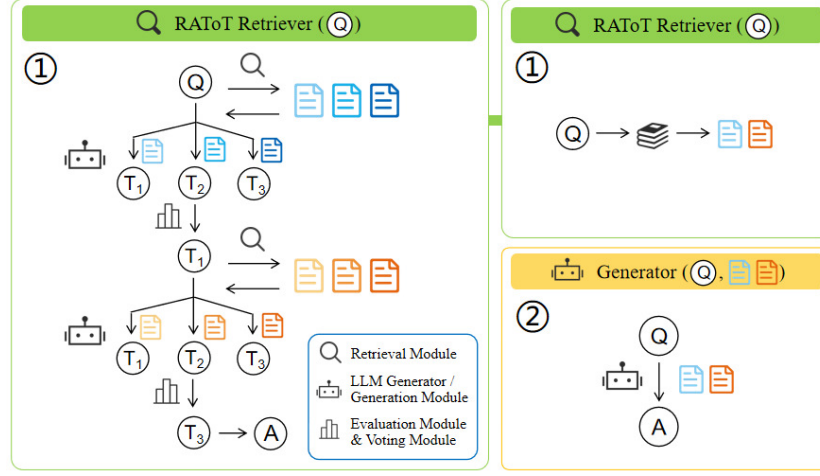


Fig. 1. Overview of **R**etrieval **A**ugmented **T**ree of **T**houghts (RAToT). Left side is the detailed structure of the right-hand RAToT Retriever. Given an open-domain question Q , RAToT retrieves documents through an iterative process. Each iterative step generates K possible following reasoning steps T_1, T_2, T_3 , et al. based on the K newly retrieved documents separately, and only retains the locally optimal T_i (for further retrieval) as well as its corresponding document. Generator then answers the question Q using all documents returned by RAToT.

is prompted to generate the temporary following inference process based on one separated document from the last K documents retrieved by Retrieval Module and the precedent returned documents. The first natural language sentence of newly generated reasoning process is used as the next reasoning step, while the rest will be re-generated in the following iterative process.

Evaluation Module Evaluation Module accepts the candidate set of next reasoning steps returned by Generation Module as input, prompts LLM to perform evaluation and pick out the most promising one by generating the evaluation result “The best CHOICE is i ”, where i is the index of the chosen reasoning step. In order to enhance the reliability of the evaluation, Evaluation Module samples this process N times and returns a candidate set of evaluation results.

Majority Voting Module Voting Module selects the most consistent evaluation result and returns the chosen reasoning step for following retrieval. The corresponding knowledge document is also retained as a search result of Retriever.

Generator Generator accepts the original question and all the search results of Retriever, uses LLM to generate a retrieval augmented chain of thoughts and answer the question. The CoT prompting template is the same as that of Generation Module.

4 Experiments

4.1 Experimental settings

We evaluate the performance of LLM QA using RAtOT retriever on 3 open-domain multi-step QA dataset, HotpotQA, 2WikiMultihopQA and MuSiQue, comparing with IRCOT under same experimental settings. For each dataset, we use the same wikipedia corpus as Trivedi et al. [18], as well as 100 randomly sampled questions as the test set.

Retrieval Algorithm We also use BM-25 for Retrieval Module. Each time called, Retrieval Module returns $K=6$ documents.

LLM Generator we use GPT-3 (gpt-3.5-turbo-instruct) as the backbone LLM of Generation Module, Evaluation Module and Generator. The corresponding backbone LLM of IRCOT is codex-davinci-002, which has been deprecated. The main difference lies in the context limit, gpt-3.5-turbo-instruct allows 4k token while codex-davinci-002 allows 8k, which means nearly double in-context demonstrations. For fair comparison, we repeat the experiment results of IRCOT on the aforementioned 3 datasets using the same backbone LLM (gpt-3.5-turbo-instruct) and test set (100 question for each dataset) as RAtOT.

Self-Consistency Majority Voting In order to balance the reliability and creativeness of LLM, we sample the evaluation process for $N=6$ times with temperature=0.7 for LLM generation.

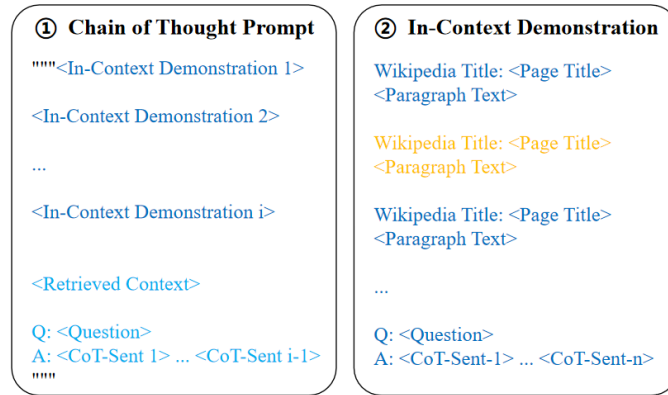


Fig. 2. Few-shot CoT prompting and in-context demonstration templates. Each in-context demonstration is a retrieval augmented CoT reasoning process for multi-step question answering. The knowledge paragraphs include the relevant paragraphs (blue) and 2 distractor paragraphs (yellow).

CoT Prompting We use the same few-shot CoT prompting template and in-context demonstrations as Trivedi et al. [18] for Generation Module and Generator, as shown in Fig. 2. We report the mean and standard deviation of experiment results using 3 demonstration sets created by Trivedi et al. Each set contains 15 in-context demonstrations of retrieval augmented reasoning QA. Due to the limitation of context window, some of them may be discarded when prompting LLM.

Evaluation Prompting We use the instruction prompting template shown in Fig. 3 for Evaluation Module, which guides LLM to evaluate and pick out the locally optimal alternative reasoning step.

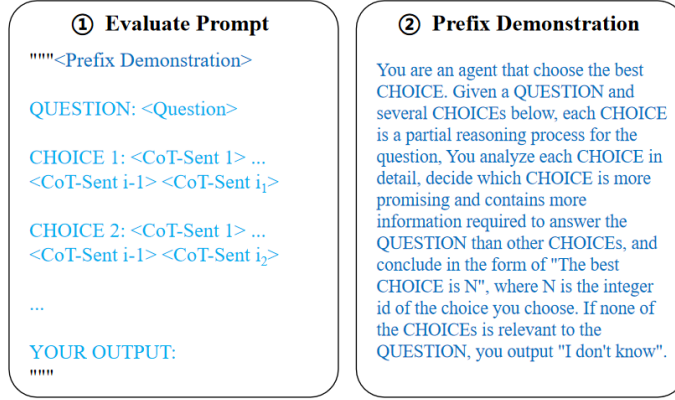


Fig. 3. Evaluation prompting template. The prefix demonstration is inspired by Yao et al. [23].

4.2 Results

Table 1. Retrieval F1 for RATOt and IRCOT using GPT-3.

	MuSiQue	2WikiMQA	HotpotQA
IRCOT	0.206 $\pm$ 0.011	0.322 $\pm$ 0.004	0.245 $\pm$ 0.003
RATOt	0.316$\pm$0.008	0.452$\pm$0.02	0.435$\pm$0.017

Table 2. Answer EM for RATOt and IRCOT using GPT-3.

	MuSiQue	2WikiMQA	HotpotQA
IRCOT	0.097 $\pm$ 0.038	0.433 $\pm$ 0.072	0.38 $\pm$ 0.035
RATOt	0.167$\pm$0.035	0.433$\pm$0.055	0.39$\pm$0.017

Table 3. Answer F1 for RAtot and IRCoT using GPT-3.

	MuSiQue	2WikiMQA	HotpotQA
IRCoT	0.203 $\pm$ 0.037	0.589$\pm$0.088	0.537 $\pm$ 0.039
RAtot	0.276$\pm$0.014	0.567 $\pm$ 0.058	0.554$\pm$0.006

RAtot can retrieve knowledge documents with less noise. Table 1 presents a comparison of RAtot and IRCoT Retriever. RAtot outperforms IRCoT by over 10 retrieval f1 point on each dataset, which indicates that the ratio of relevant documents to noisy ones in retrieved documents using RAtot is higher than using IRCoT, thus overall RAtot performs retrieval with higher quality through making locally optimal decision for each iterative step.

RAtot can enhance complex reasoning QA. Table 2 and Table 3 compare the QA performance of retrieval augmented LLM Generator using RAtot and IRCoT as retriever. On the one hand, RAtot outperforms IRCoT by 7.3 answer F1 point and 7 EM point on MuSiQue, which shows higher factuality and less hallucinations. Compared with other 2 datasets, MuSiQue consists of more complex reasoning questions and is harder to cheat via reasoning shortcuts [17]. The performance on MuSiQue indicates that retrieving selectively and flexibly using RAtot can augment LLM reasoning effectively. On the other hand, RAtot does not significantly enhance the QA performance on HotpotQA and fails to improve the performance on 2WikiMultihopQA. We mainly attribute this flaw to the erroneous discarding of relevant documents. Specifically, RAtot selects locally optimal reasoning approach and its corresponding document using Evaluation Module in each iterative step, but LLM sometimes outputs seemingly correct reasoning approaches without realizing they have been misled by noisy information, and there may be not obvious difference between these much confusing reasoning approaches and the correct one. In this situation, Evaluation Module may pick out a fallacious reasoning path and corresponding noisy document, which results in a decline of retrieval quality and fewer enhancements of LLM reasoning QA in some degree.

5 Conclusion

We propose RAtot, a iterative retrieval framework which retrieves knowledge documents flexibly and wisely by generating a step by step reasoning process, each step of which is locally optimal. In general scenarios where a large amount of knowledge documents are retrieved to find the useful one, RAtot is a promising approach to retrieve documents with less noise and answer knowledge-intensive open-domain questions with less hallucination.

RAtot also has several limitations. RAtot selects optimal reasoning steps and its corresponding retrieved documents at a risk of discarding relevant documents sometimes and thus may miss useful information. Besides, we observe that LLM generates unexpectedly convincing and fluent reasoning processes based

on noisy documents occasionally, which makes it some difficult to distinguish correct reasoning process from them automatically. This phenomenon may be relevant to the over-confidence of LLMs [27]. Future works could develop more efficient evaluation mechanism to further improve the robustness of RAToT and extend the reasoning framework to more complex graph structure to enhance the reasoning capability of LLM QA system.

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